



THE UNIVERSITY OF
CHICAGO

MASTERS IN
COMPUTATIONAL
SOCIAL SCIENCE
THE UNIVERSITY OF CHICAGO

A large, semi-transparent image of Po the panda from the movie Kung Fu Panda. He is wearing his signature conical straw hat and a red and yellow scarf. He has a confident, slightly smug expression. The background of the image is a soft-focus view of a traditional Chinese village with tiled roofs.

MACS 30111

M7: Pandas

Announcements

▶ **Final exam schedule**

- ▶ Tues, Dec 9^h, 12:30 PM-2:30 PM
- ▶ Take-home: open Friday through Weds
 - ▶ Timing: 36 hours (same as midterm)

▶ **Review**

- ▶ Textbook
- ▶ Team tutorial
- ▶ Short exercise
- ▶ Programming assignment
- ▶ Extra exercise

Key commands

- ↳ Loading: `read_csv()`
- ↳ Summarizing:
 - ↳ `data.head()`
 - ↳ `data.tail()`
- ↳ Selecting:
 - ↳ By index: `iloc` (e.x. `trees.iloc[:, 1]`)
 - ↳ By name: `loc` (e.x. `trees.loc[:, "block_id"]`)

Motivating example

- ▶ Data exploration for the 2015 New York City Street Tree Survey
- ▶ Data: <https://data.cityofnewyork.us/Environment/2015-Street-Tree-Census-Tree-Data/uvpi-gqnh>
 - ▶ CSV, 683,789 lines

1. How many different species are planted as street trees in New York?
2. What are the five most common street tree species in New York?
3. What is the most common street tree species in Brooklyn?
4. What percentage of the street trees in Queens are dead or in poor health?
5. How does street tree health differ by borough?

https://github.com/computer-science-with-applications/examples/tree/main/working_with_data/pandas

Coding practice: 4.3.1

Selecting rows and columns

Note: depending on data, might have 'boroname'

```
In [19]: trees.iloc[:5][["health", "borough"]]
```

```
Out[19]:
```

	health	borough
0	Fair	Queens
1	Fair	Queens
2	Good	Brooklyn
3	Good	Brooklyn
4	Good	Brooklyn

```
In [20]: trees.loc[[180683, 200540, 204026], ["health",  
...: "borough"]]
```

```
Out[20]:
```

	health	borough
180683	Good	Brooklyn
200540	NaN	Staten Island
204026	Good	Staten Island

Filtering

- Boolean masks
- Logical operations

```
filter = (trees.borough == "Queens") & ((trees.status == "Dead") | (trees.health == "Poor"))  
trees[filter]
```

- Remove rows where trees.status is Dead
- Select rows where trees.status is not Dead

```
trees[trees.status != "Dead"]
```

```
0    False  
1    False  
2    False  
3    False  
4    False
```

```
...  
683783  False  
683784  False  
683785  False  
683786  False  
683787  False
```

```
Name: status, Length: 683788, dtype:  
bool
```

Common Summary Statistics

DataFrame.count

Count number of non-NA/null observations.

DataFrame.max

Maximum of the values in the object.

DataFrame.min

Minimum of the values in the object.

DataFrame.mean

Mean of the values.

DataFrame.std

Standard deviation of the observations.

DataFrame.select_dtypes

Subset of a DataFrame including/excluding columns based on their dtype.

Summary statistics for each column: `df.describe()`

Summary of categorical values: `df.value_counts()`

Pairwise correlation of columns: `df.corr()`

```
In : trees.count()
```

```
Out :
```

block_id	683788
created_at	683788
tree_dbh	683788
stump_diam	683788
curb_loc	683788
status	683788
health	652172
spc_latin	652169
spc_common	652169

```
In : trees.status.value_counts()
```

```
Out :
```

status	
Alive	652173
Stump	17654
Dead	13961

Name: count, dtype: int64

Reference: <https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.describe.html>

Using functions

What does this do/mean? Why and how would we do this?

```
for col_name in ["boroname", "health",  
"spc_common", "status"]:  
    trees[col_name] =  
    trees[col_name].astype("category")
```

Using functions

↳ What does this do/mean? Why and how would we do this?

```
def tree_health_by_boro(trees):  
    combined_status = trees.status.where(trees.status != "Alive", trees.health)  
    num_per_boro = trees.groupby("borough").size()  
    combined_per_boro = trees.groupby(["borough", combined_status]).size()  
    pct_per_boro = combined_per_boro/num_per_boro*100.0  
    pct_per_boro_df = pct_per_boro.unstack()  
  
    return pct_per_boro_df[["Good", "Fair", "Poor", "Dead", "Stump"]]
```

```
tree_health_by_boro(trees)
```

New variables: creating new from old

- ↳ Start with continuous-y variables and can reshape - maybe you want or need a categorical, maybe you are looking to reshape
- ↳ Categorical
- ↳ Cut (and qcut)
- ↳ Where (np.where)
- ↳ Apply
- ↳ Map
- ↳ Bonus: lambda functions

df.where

```
└ cond = trees_life['Average_lifespan'] > 150
└ trees_life["lifespan_short"] =
  trees_life['Average_lifespan'].where(cond, other =
    "short lived")
└ trees_life["lifespan_short"].value_counts()
```

Pd.cut

- ↳ This is one option to make more variables - I like it because you can slice the variable in different ways. Basically, it's for when you have a continuous-y variable and you want to do some calculating or plotting by groups.

Cutting:

↳ Cut the range into equal slices:

```
trees_life["lifespan_cat_split"] = pd.cut(trees_life['Average_lifespan'],  
bins=4,  
labels=['Short', 'Medium', 'Long', 'ExtraLong'])
```

↳ Cut the distribution into equal slices:

```
trees_life["lifespan_cat_eq"] = pd.qcut(trees_life['Average_lifespan'],  
q=4,  
labels=['Short', 'Medium', 'Long', 'ExtraLong'])
```

↳ DIY where you set the cut points:

```
trees_life["lifespan_cat"] = pd.cut(trees_life['Average_lifespan'],  
bins=[0, 90, 200, 300, 600], right = True,  
labels=['Short', 'Medium', 'Long', 'ExtraLong'])
```

Pivot example

```
filter_col = [col for col in trees_life if col.startswith('lifespan_cat_eq')]
```

```
pd.melt(trees_life, value_vars=filter_col,  
id_vars= ["Common Name", "Scientific Name"],  
var_name='lifespan_cat')
```

```
pd.wide_to_long(trees_life, stubnames="lifespan_cat_eq",  
i="Common Name",  
j="lifespan_cat", suffix='\\w+')
```

```
trees_life.pivot(index='Common Name', columns='lifespan_cat',  
values='Average_lifespan')
```

```
(bonus) pd.get_dummies(trees_life, columns=['lifespan_cat'], dtype=int)
```

Combining DataFrames

- **concat()**
 - perform concatenation along an axis
 - while performing set logic of the indexes on other axes
 - make a full copy of the data
- **merge()**
 - Standard database join operations between DataFrame or Series objects
- **join()**
 - join on index
 - combine columns of two differently indexed DataFrames into a single one

Reference: https://pandas.pydata.org/docs/user_guide/merging.html

Toy examples

↳ Consider the following three dataframes:

↳ `df1 = pd.DataFrame([[ 'a', 1], [ 'b', 2]], columns=[ 'letter', 'number'])`

↳ `df2 = pd.DataFrame([[ 'a', 1], [ 'c', 4]], columns=[ 'letter', 'number'])`

↳ `df3 = pd.DataFrame([[ 'b', 1], [ 'd', 2]], columns=[ 'entry', 'number'])`

`pd.concat([df1,df2])`

`pd.concat([df1,df3])`

`pd.merge(df1,df3, on = "number")`

`df3.set_index("number").join(df1.set_index("number"))`

`pd.merge(df1,df2, on = "number")` ## Play around with this and 'how' options

Join vs merge

- ↳ Merge is the ‘big picture’ for things
 - ↳ Can have more freedom with indices / how you merge
- ↳ ‘join’ is like a subset of merge - think of it like a merge based on the index. Can be faster than merge.

Merge: syntax

↳ **`DataFrame.merge(right, how='inner', on=None, left_on=None, right_on=None, left_index=False, right_index=False, sort=False, suffixes=('_x', '_y'), copy=None, indicator=False, validate=None)`**

Parameters:

- **how**{'left', 'right', 'outer', 'inner', 'cross'}, default 'inner' Type of merge to be performed.
 - left: use only keys from left frame, similar to a SQL left outer join; preserve key order.
 - right: use only keys from right frame, similar to a SQL right outer join; preserve key order.
 - outer: use union of keys from both frames, similar to a SQL full outer join; sort keys lexicographically.
 - inner: use intersection of keys from both frames, similar to a SQL inner join; preserve the order of the left keys.
 - cross: creates the cartesian product from both frames, preserves the order of the left keys.
- **On** *label or list* Column or index level names to join on. These must be found in both DataFrames. If *on* is None and not merging on indexes then this defaults to the intersection of the columns in both DataFrames.

Syntax: is it `df.merge` or `pd.merge`?

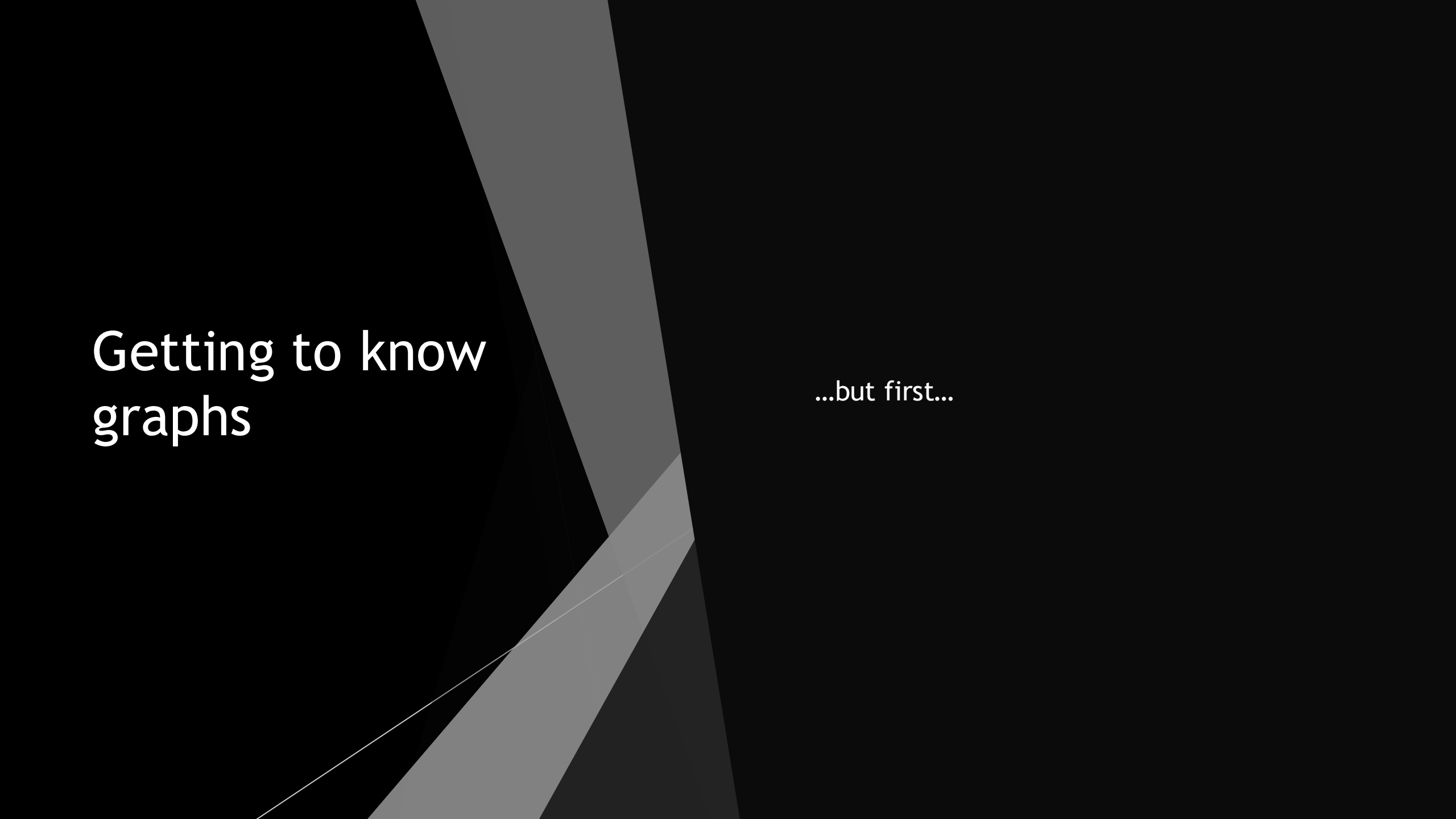
Similar across circumstances but just depends on how you want to call it...when you look at the documentation, you'll see this trend consistently.

If you call `pd.function`, you will need to specify the data frame(s).

If you `df.function`, you no longer need to specify the data frame.

Merge / Join hints

- ↳ Are you bringing together based on index (use join) or something else (use merge)
- ↳ Probably want to have named index but could have circumstances where it doesn't make sense for your research question



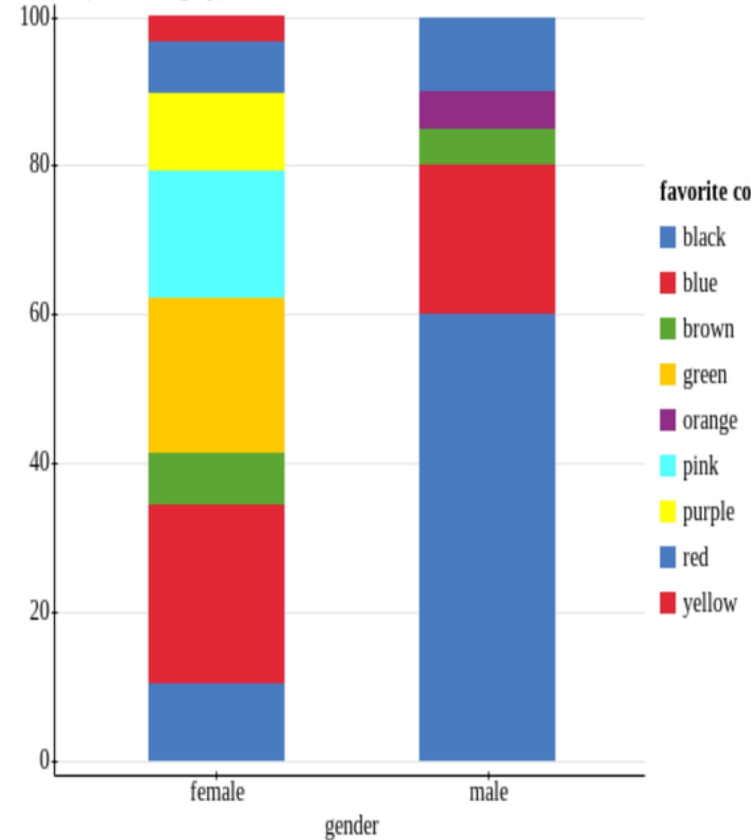
Getting to know graphs

...but first...

Q9 Could you provide us with your Postal Code?

Result 2: segmented par graph

Percent (within category)



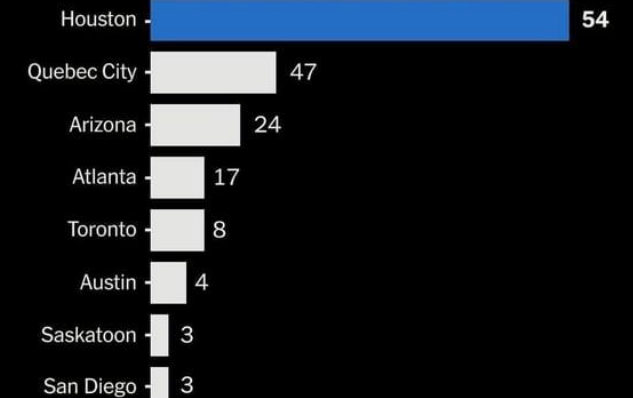
Question options

- Dartmouth, NS, B2V0A8
- Lower Sackville, NS, B4E0J5
- Bedford, NS, B4A4H1
- Dartmouth, NS, B3K2Z4
- Dartmouth, NS, B3A3N2
- Dartmouth, NS, B2V2M7
- Dartmouth, NS, B3A3J1
- Dartmouth, NS, B2Y4L2
- Halifax, NS, B3K3L5
- Halifax, NS, B3H3Z
- Dartmouth, NS, B2X1G2
- Halifax, NS, B3K2E3
- Fall River, NS, B2T1A4
- Halifax, NS, B3K4Z
- Halifax, NS, B3M4N6
- Dartmouth, NS, B3A4X8
- Halifax, NS, B3M0A1
- Dartmouth, NS, B2X1G2
- Cole Harbour, NS, B2V0A1
- Halifax, NS, B3K1T6
- Lower Sackville, NS, B4F1H1
- Halifax, NS, B3K1T6

Anonymous NHL Player Poll

Where would you like to see a new NHL team play?

175 Votes



Teams receiving two votes: Kansas City, Helsinki, Oklahoma City and Miami
Teams receiving a single vote: Boise, Dubai, Green Bay, Halifax, Jackson Hole, Milwaukee and Orlando.

Source: The Athletic NHL staff (survey of players) • Poll open from Sept. 27 to Nov. 10, 2024



<https://www.reddit.com/r/dataisugly/>

Starting with a summary:

```
def tree_health_by_boro(trees):  
    combined_status = trees.status.where(trees.status != "Alive", trees.health)  
    num_per_boro = trees.groupby("borough").size()  
    combined_per_boro = trees.groupby(["borough", combined_status]).size()  
    pct_per_boro = combined_per_boro/num_per_boro*100.0  
    pct_per_boro_df = pct_per_boro.unstack()  
  
    return pct_per_boro_df[["Good", "Fair", "Poor", "Dead", "Stump"]]  
  
tree_health_by_boro(trees)
```

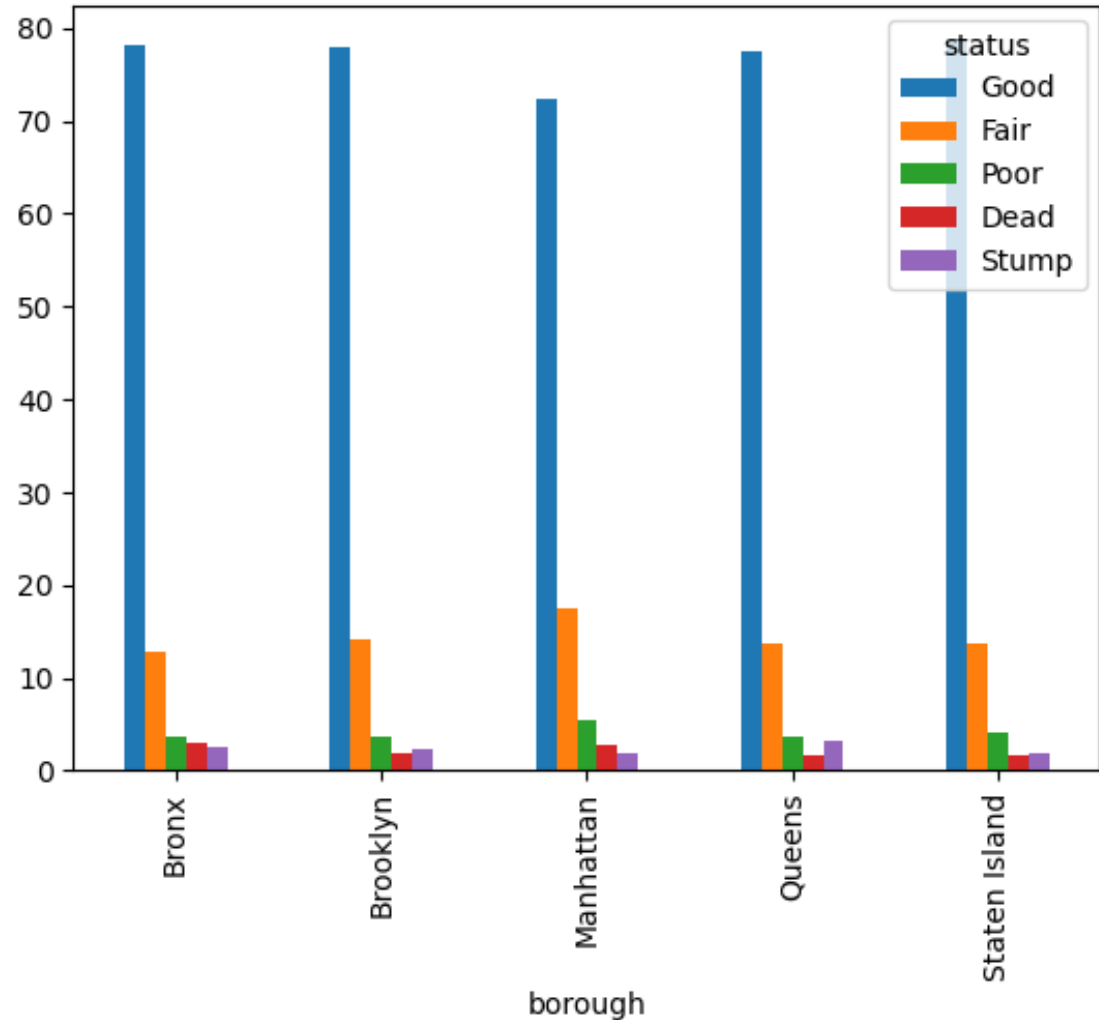
status	Good	Fair	Poor	Dead	Stump
borough					
Bronx	78.169783	12.777719	3.632501	2.969379	2.450618
Brooklyn	77.956829	14.142126	3.643122	1.872042	2.385881
Manhattan	72.387387	17.516775	5.516409	2.754383	1.825046
Queens	77.432539	13.789209	3.758516	1.772094	3.247642
Staten Island	78.494654	13.801060	4.024003	1.775575	1.903758

output

Plotting with Pandas

Using the plot method:

```
↓ tree_health_by_boro(trees).plot.bar()
```



Reference:

<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.plot.html>

Plotting with matplotlib

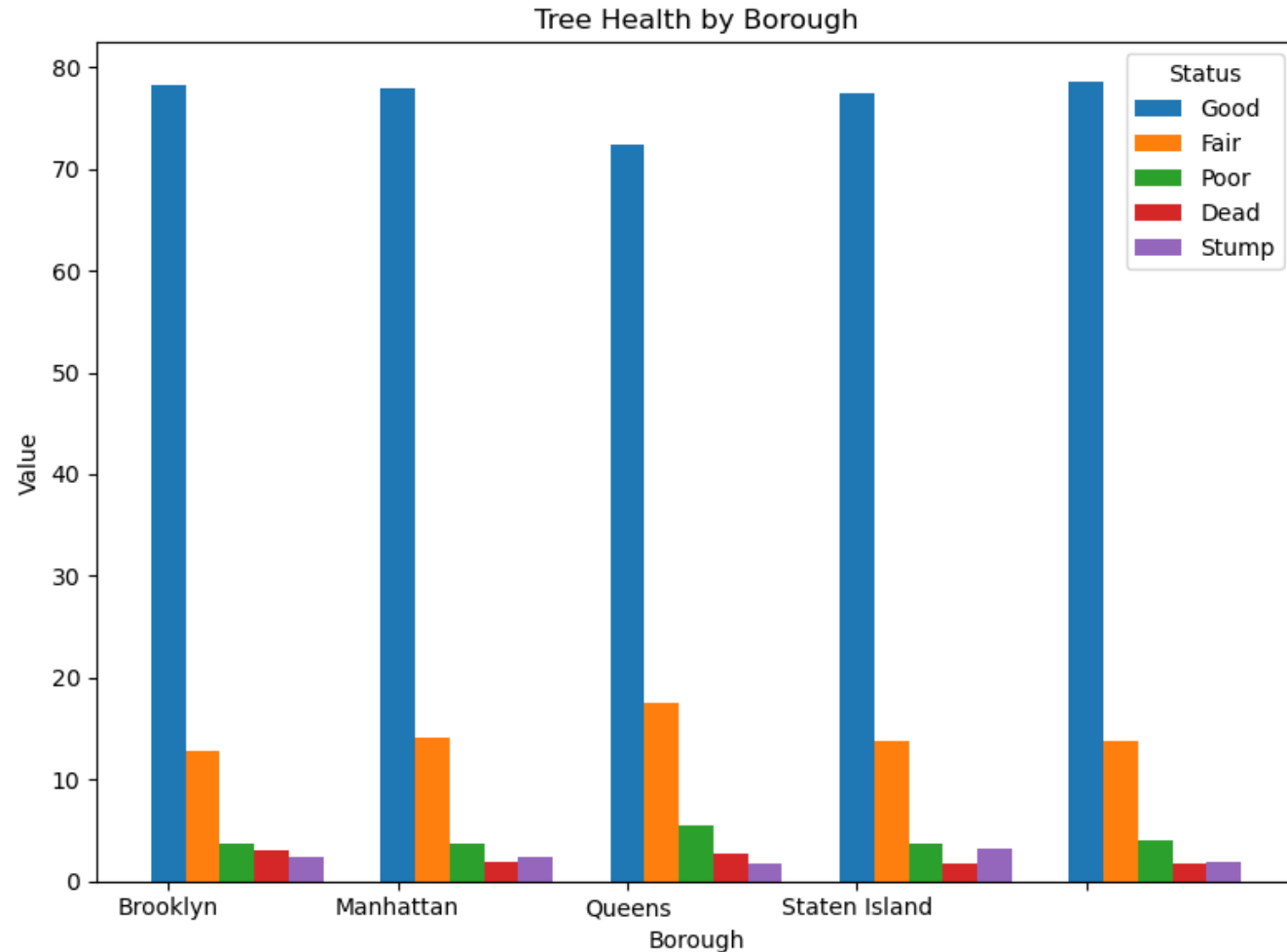
```
import matplotlib.pyplot as plt
import numpy as np
df = pd.DataFrame(tree_health_by_boro(trees)).reset_index()

# Group by borough and status, summing the counts
df_melted = df.melt(id_vars="borough", var_name="status", value_name="value")
fig, ax = plt.subplots(figsize=(8, 6))
bar_width = 0.15
index = np.arange(len(df['borough']))

# Plot bars for each health status
for i, status in enumerate(df_melted['status'].unique()):
    status_values = df_melted[df_melted['status'] == status]['value']
    ax.bar(index + i * bar_width, status_values, bar_width, label=status)

# Customize plot
ax.set_xticklabels(df['borough'])
ax.set_xlabel('Borough')
ax.set_ylabel('Value')
ax.set_title('Tree Health by Borough')
ax.legend(title="Status")

# Show plot
plt.tight_layout()
plt.show()
```



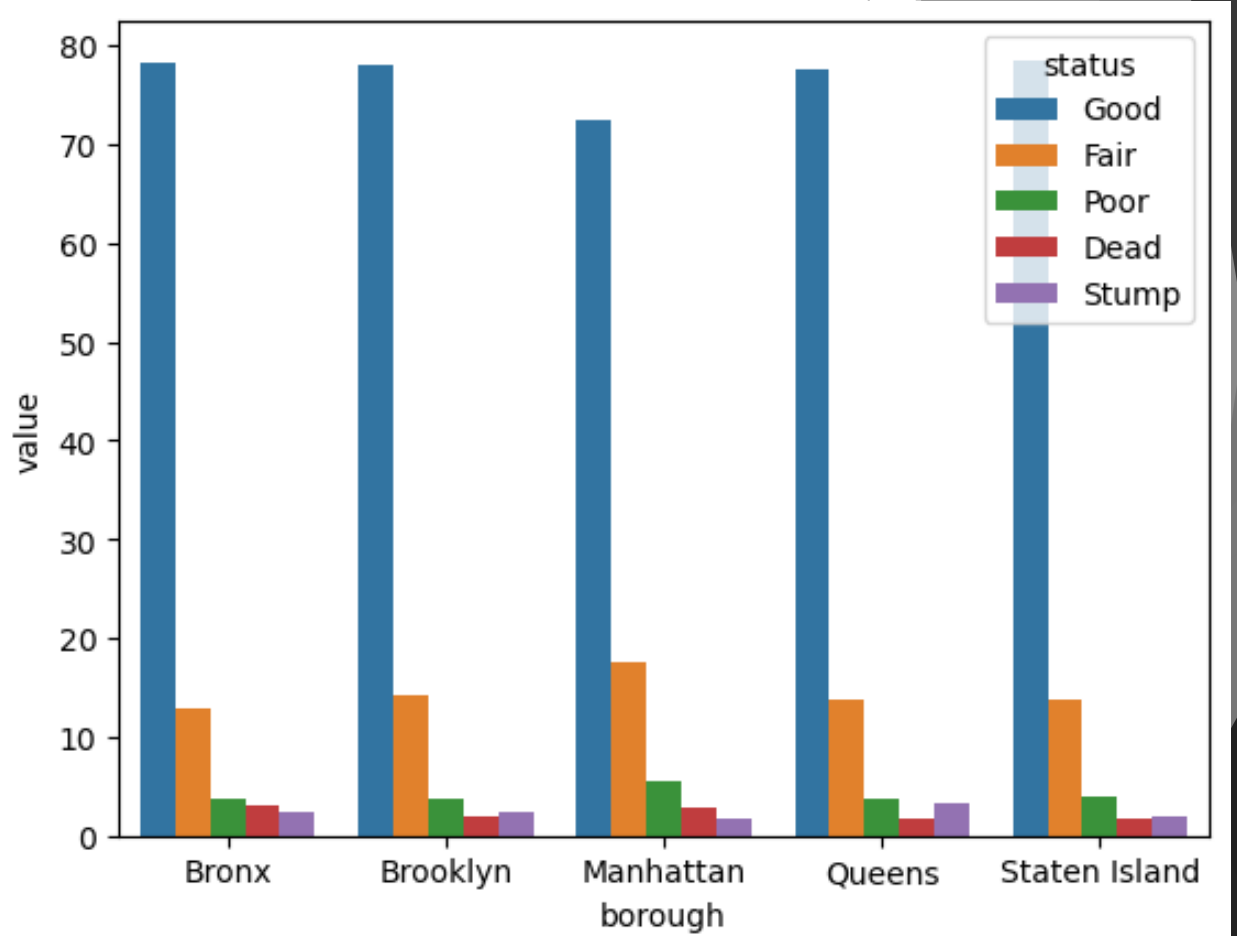
Plotting with Seaborn

Seaborn

```
df2 = df.reset_index()
```

```
df2 = df2.melt(id_vars = "borough")
```

```
sns.barplot(df2, x = "borough", y = "value", hue =  
"status")
```



Which to use?

- ↳ Probably not matplotlib...
- ↳ Usually, quick and dirty, proof-of-concept: basic plot methods but nice and pretty: **seaborn**
- ↳ PRACTICE PRACTICE PRACTICE

Misc: other POLARS



Seems cool but pandas is probably better for what you need right now

<https://pola.rs/>

RECAP

- ▶ Pandas is going to be HUGE (ditto NumPy!)
- ▶ Think about what you need
- ▶ Sometimes it is MUCH faster to puzzle through and sketch before trying to do something - often there is a simpler path
- ▶ Think about what you are trying to do and what it will look like.
 - ▶ Do you need a new column?
 - ▶ Are you summarizing data?
 - ▶ Is this a 'permanent' or 'temporary' alteration?
- ▶ Graphing: google search **IMAGES**